CAMPUSAI

ARTIFICIAL INTELLIGENCE (AI) – ONE-PAGE SAMPLE NOTES

1. Introduction

Artificial Intelligence (AI) is a branch of computer science that develops systems able to perform tasks that normally require human intelligence, such as learning, reasoning, problem-solving and understanding language.

2. Important Concepts

- **Intelligent Agent:** A system that observes its environment and takes actions to achieve a goal.
- **Search:** Exploring possible choices to find a solution to a problem.
- **Knowledge Representation:** Organising facts and relationships so a computer can use them for reasoning.
- **Machine Learning (ML):** A part of AI in which computers learn patterns from data.
- **Natural Language Processing (NLP):** Helps computers understand and generate human language.
- **Computer Vision:** Helps computers analyse images and videos.

3. Types of Machine Learning

- **Supervised learning:** Learns from labelled examples.
- **Unsupervised learning:** Finds patterns in data without labels.
- **Reinforcement learning:** Learns through rewards and penalties.

4. Applications of AI

Chatbots and virtual assistants • Medical image analysis • Recommendation systems • Fraud detection • Robotics • Language translation

5. Example

A college chatbot can answer students' common questions about class timings, exam schedules and campus services using verified college information.

6. Quick Revision Questions

1 Define Artificial Intelligence.

2 What is an intelligent agent?

3 Differentiate supervised and unsupervised learning.

4 Write any four applications of AI.

Note: These are introductory sample notes. Refer to your course syllabus and class materials for complete coverage.